13

Effective HACCP training

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Reading Scientific Services Ltd (RSSL) is a leading supplier of outsourced scientific services, including consultancy, training, research and analysis. A wide range of HACCP training courses are provided through RSSL's training company, Food Industry Training – Reading (FITR), a joint venture with the Department of Food Biosciences at the University of Reading.

13.1 Introduction

> HACCP is carried out by people. If the people are not properly experienced and trained then the resulting HACCP system is likely to be ineffective and unsound. (Mortimore and Wallace 1998)

Effective training is critical to the successful implementation of HACCP. Training in HACCP has two main roles:

1. It helps to develop awareness and motivation in the workforce for food safety management.
2. It provides technical and practical knowledge enabling trainees to participate in HACCP development and implementation.

Failure in either role will leave staff unwilling or unable to meet their HACCP responsibilities. There are a number of reasons that training can be less effective than planned in meeting these objectives. These include:

- motivation: initial resistance to training among trainees
- understanding: the failure to appreciate the kind of training required

- targeting: training the wrong people and/or providing the wrong level of training at the wrong time
- follow-up: the lack of follow-up from training
- quality: inconsistent and/or poor quality in training provision
- resources: the lack of resources for effective training, particularly for small and medium sized enterprises (SMEs)
- coverage: not training enough people.

This chapter looks at how to design and implement effective HACCP training, overcoming these potential problems to ensure achievement of the desired learning outcomes.

13.2 Training needs analysis

It is important to consider the differing levels of HACCP training needed within an organisation. Trainee groups will include:

- senior management
- the HACCP team and team leader
- CCP monitors and their supervisors
- other production staff.

The roles that these people will play in HACCP development and implementation will define their learning needs and will help in the development of training and learning objectives for each group.

Once the senior management team has committed itself wholeheartedly to implementing a HACCP system, an evaluation of the training requirements of all these groups will be an essential part of the initial planning for HACCP application. An effective evaluation will need to take account of the learning needs of the groups outlined above, from HACCP team leader to the workforce as a whole, and the appropriate timing for such training. It may be appropriate for a member of the HACCP team to take responsibility for the training needs evaluation, working in conjunction with the organisation's own human resources function to produce a detailed training plan as part of the HACCP implementation plan. It is important at this stage to review the comprehensiveness and effectiveness of existing training in food hygiene since this will be an essential prerequisite to specific training in HACCP implementation.

In addition to training in HACCP techniques, training in associated skills needed to develop and implement a HACCP system may be required, e.g. project management skills and leadership skills. Possible training objectives for different groups of staff are outlined in Table 13.1. The resulting HACCP training programme and its outcomes will need to be documented as part of the overall HACCP implementation plan.

Table 13.1 Possible training objectives for different groups of staff

Group	Training objective
Senior management	1. Understand the general principles of HACCP and how they relate to the food business. 2. Demonstrate an understanding of the training and knowledge requirements for HACCP team members and the workforce as a whole. 3. Demonstrate an understanding of the links between HACCP and other quality management techniques and programmes and how a combined product management system can be developed. 4. Understand the need to plan the HACCP system and develop a practical timetable for HACCP application in the whole operation.
HACCP team leaders	*HACCP system and its management*[a] 1. Demonstrate an up-to-date general knowledge of HACCP. 2. Explain how a HACCP system supports national and international standards, trade and legislative requirements. 3. Describe the nature of prerequisite programmes and their relationship with HACCP. 4. Demonstrate the ability to plan an effective HACCP system. 5. Demonstrate a knowledge of how to lead a HACCP team. 6. Demonstrate an understanding of the practical application of HACCP principles. 7. Demonstrate the ability to design, implement and manage appropriate programmes for verification and maintenance of HACCP systems. 8. Explain the methods to be used for the effective implementation of HACCP. *Additional topics* 1. Demonstrate an understanding of the nature of hazards and how they are manifested in food products/operations and give relevant examples. 2. Demonstrate an understanding of the intrinsic factors governing the safety of product formulations and methods that can be used to assess safety of new products. 3. Carry out the steps to identify significant hazards relevant to the operation and determine effective control measures, i.e. assessment of risk (likelihood of occurrence and severity). 4. Demonstrate an understanding of the training and knowledge requirements for HACCP team members and the workforce as a whole. 5. Develop appropriate training programmes for CCP monitoring personnel. 6. Demonstrate an understanding of the links between HACCP and other quality management techniques and how a combined product management system can be developed.

Table 13.1 (continued)

Group	Training objective
HACCP team members	*HACCP system*[b] 1. Justify the need for a HACCP system. 2. Show how the legal obligations on food business proprietors to analyse food hazards and identify critical steps in the business activities should be met in their appropriate industries. 3. List and explain the importance of the principles of HACCP. 4. Describe the method by which hazard analysis may be carried out and appropriate control measures ascertained to assess the practical problems. 5. Identify critical control points including critical limits to ensure their control. 6. Develop suitable monitoring procedures for critical points and explain the importance of corrective action procedures. 7. Verify the HACCP system by the use of appropriate measures. 8. Carry out the steps to introduce and manage a fully operational HACCP system. *Additional topics* 1. Demonstrate an understanding of the nature of hazards and how they are manifested in food products/operations and give relevant examples. 2. Demonstrate an understanding of the intrinsic factors governing the safety of product formulations and methods that can be used to assess safety of new products. 3. Carry out the steps to identify significant hazards relevant to the operation and determine effective control measures, i.e. assessment of risk (likelihood of occurrence and severity). 4. Develop appropriate training programmes for CCP monitoring personnel.
CCP monitors	1. Understand the general principles of HACCP and how they relate to the food handler's role. 2. Perform CCP monitoring tasks, record results and initiate appropriate actions.
Auditors of HACCP systems	*HACCP and regulatory auditing*[c] 1. Provide up-to-date general knowledge of HACCP and its relationship with national and international standards, trade requirements and legislative requirements. 2. Examine the role of good hygiene practices as a foundation for HACCP-based food safety management systems. 3. Provide a comprehensive revision of the application of HACCP principles for the development of HACCP-based systems for food businesses.

	4. Consider the design and management requirements associated with the application and implementation of HACCP-based food safety management systems in food businesses. 5. Enhance the skills required for the assessment of HACCP-based food safety management systems. 6. Consider the tools available to educate food business operators in the principles of HACCP and to provide advice and support during development and implementation of food safety management systems.
	Additional topics 1. Understand the need for audit preparation including the development of suitable check-lists. 2. Perform HACCP audits using sampling, questioning, observation and assessment skills. 3. Construct audit reports giving clear indication of findings and corrective action needed.
General workforce	1. Understand the general principles of HACCP and how they relate to the food handler's role.

Sources: [a] = UK Steering Group on HACCP Training Standards (1999)
[b] = Royal Institute of Public Health and Hygiene (1995a)
[c] = UK Food Standards Agency (2000)

13.3 Training and effective learning styles

Training is often seen as an activity that goes on in the 'classroom'. Indeed, much initial training does take place away from the trainee's ordinary activities in group training sessions. There are three immediate limitations to training of this kind:

1. 'Classroom' sessions are inevitably limited in time: staff may only be able to attend occasional training programmes, with the risk that the impact of such training gradually becomes diluted over time.
2. Training away from the production operations themselves may make it more difficult to relate what is learnt back to the specific operations.
3. There may also be issues of motivation in employees attending HACCP training sessions for the first time. They may be viewed initially with suspicion and cynicism, at best as an irrelevance, at worst as an unwelcome fad foisted on an unwilling workforce by management.

These issues need to be considered in designing the HACCP training programme.

The training needs of particular groups, and the kinds of training programme they require, are explored elsewhere in this chapter. Whatever group is being trained, it is important to be aware of the range of learning styles among any group. While some trainees may be capable of absorbing large amounts of data in a classroom setting, others will be less accustomed to this kind of environment and will require a range of techniques if they are to learn effectively. Some

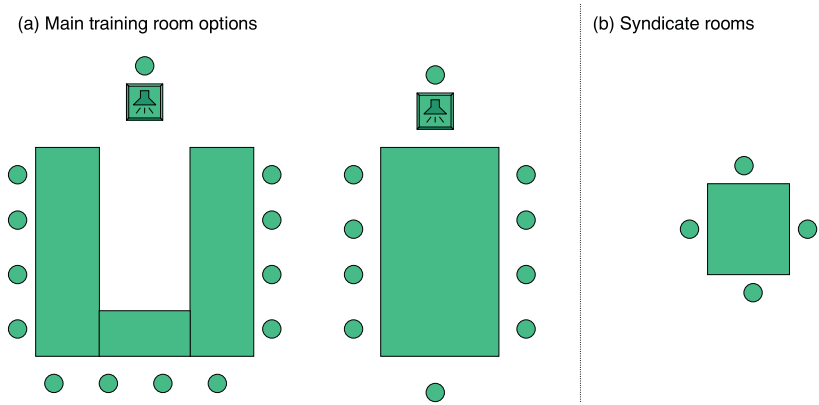


Fig. 13.1 Effective seating configurations for training.

will have negative preconceptions of training as a dull, passive and unproductive process, which will have to be overcome in the early stages of the training session. The right kind of seating arrangement designed to break down a sense of formality and encourage participation will help counteract such preconceptions and create the right kind of learning environment (Fig. 13.1). Getting trainees to introduce themselves or each other is a useful way of breaking the ice. Communicating a genuine sense of enthusiasm to the trainees is also essential, given the importance of non-verbal signals in communication.

Given that many trainees may be unused to taking in large amounts of factual data in a short space, trainers need to avoid long periods where they are talking to an essentially passive audience. Where this is necessary, the use of visual aids such as computer presentations, overheads and flip charts will help reinforce and enliven whatever a trainer says. Many trainees will also be unused to both absorbing what is being said and taking notes, so handouts summarising the key points will both allow trainees to concentrate on understanding and reinforce learning after the training session is over. This can be done by providing minimised versions of overheads used in any training session or by developing a simple training support booklet, e.g. *HACCP: Your Questions Answered* (Mortimore and Wallace 1997). Another possibility is to provide handouts in the form of laminated pocket-sized cards listing issues such as key hazards and control measures for regular consultation.

Learning is always more effective if it is made interactive. Asking questions and encouraging participation and debate encourages trainees to think actively about what they are learning. ‘Doing’ is often the best way of learning effectively, and training sessions should mix theory with practical application. Trainers can use group workshops to work through practical examples. Presenting findings to other groups can be an excellent way of focusing minds and generating a sense of involvement from members of a group. In such an environment, trainees themselves take on a training role, explaining issues and supporting other members of their group to ensure that the group as a whole succeeds.

Any initial classroom briefings must be part of a wider training programme that is seen to build on these beginnings. It is critical to follow through with either more detailed training for those involved in HACCP implementation, CCP monitors for example, or clear signs that a HACCP system is being developed and implemented in the organisation. In general, to be most effective, training should have follow-up such as action points with feedback on action taken. It may be appropriate to have some more formal assessment of training outcomes as part of such follow-up, perhaps via an exam (as a conclusion to an intensive training session for HACCP team leaders, for example) or assessment of performance in HACCP implementation by the relevant supervisor or line manager. This follow-up needs to be seen both as an integral part of the training process and as a way of assessing its effectiveness. More fundamentally, training outcomes may need to be linked to revisions to job descriptions, revised individual and departmental objectives, and changes to performance appraisal and bonus schemes.

13.4 Appointing HACCP trainers

Organisations must also decide who will do the training, i.e. whether the use of external trainers will provide the most effective training, or whether to use their own resources for some or all training. With the exception of large companies, where there may be a central pool of expertise, most organisations will be dependent on external training providers for at least the initial training of the HACCP team and team leader, given their central role in HACCP planning and implementation. There are a number of advantages to the use of *reputable* outside trainers:

- they should have expertise in HACCP principles and methodologies
- they should have the experience, personnel and materials to deliver effective training programmes quickly and efficiently
- in some cases, such training can lead to a formal qualification generally recognised in the industry, which will support verification.

These advantages may also be seen in internal training where the company has appropriate in-house resources.

There may also be a number of disadvantages in using outside trainers:

- the quality of some training provision may be variable
- the cost of using external providers may be substantial to some businesses
- there is potential for reliance on generic training which does not relate to the particular circumstances of the organisation.

Although some HACCP training is of very high quality, there is considerable variability in the standard of training available, given the lack of regulation and standardisation in this area. In some cases, while HACCP trainers have the necessary expertise, they may lack general training skills and experience. In

others, providers may be experienced trainers but lack expertise in HACCP. Some early HACCP training has been bolted onto existing hygiene courses, without the necessary conceptual leap from hygiene management to hazard analysis, risk assessment and control (Mortimore and Smith 1998).

There have been a number of initiatives to resolve this problem. In New Zealand, for example, the meat and seafood industries, in consultation with the Ministry of Agriculture, have developed their own industry-specific HACCP competency qualifications covering such areas as coordination of the development of a HACCP plan, and supervision of a HACCP system in the workplace (see Chapter 9). The former is appropriate to HACCP team leaders and auditors, and the latter to supervisory staff responsible for the performance of CCP monitoring staff. In the UK, a group of HACCP practitioners, trainers, regulators and enforcers, the UK Steering Group on HACCP Training Standards, has sought to achieve more consistency in HACCP training by developing introductory and advanced HACCP training standards (RIPHH 1995a, UK Steering Group 1999). These standards have been designed as a benchmark against which to assess training courses for HACCP team leaders (the advanced standard) and HACCP team members (the introductory standard). The standards have been used by awarding bodies such as the Royal Institute of Public Health and Hygiene as a basis to certify the content of HACCP training courses, offer examinations as verification of trainee achievement, and to approve and register trainers. In the USA, the International HACCP Alliance provides a similar accreditation programme for HACCP training courses against detailed training objectives (International HACCP Alliance 1996).

Familiarity with these standards will help those responsible for training within organisations implementing HACCP systems to assess their training requirements as well as the content of courses available from external providers.

The problem of external training being generic and not specific to the operation can be dealt with in a number of ways:

- By early liaison between provider and client organisation on training course design to include, for example, material on the organisation's own processes.
- By approaching an organisation simultaneously for training and consultancy. In the UK, for example, MAFF has funded a combined package for retail butchers consisting of a two-day training course followed by free consultancy designed to assist the actual process of HACCP implementation. Another example might involve initial training of the HACCP team and then on-site facilitation of the HACCP process by an experienced consultant.

It is important to note, however, that sending one person on an external open course of one to two days' duration is unlikely to give them enough knowledge or experience for them to be expected to come back and manage the development and implementation of HACCP without further training/support.

The question of whether to use an external training agency for the training of CCP monitors and production personnel is complicated in part by the greater scale and cost involved in training this larger group. One solution is to

involve an external agency in training in-house staff who then carry out general training of production personnel. In this example trainers may be selected from supervisory staff who have the relevant production experience and authority both to absorb and communicate HACCP principles and skills effectively. Delegating training in this way has the potential advantage that the trainer may well know the trainees and the relevant production operations, understands trainees' needs and concerns, and has their trust. However, if in-house staff are used to train CCP monitors and other staff, they need to have the right training competencies: enthusiasm and motivational abilities, training and communication skills, the ability to manage groups of trainees, as well as an in-depth knowledge of the HACCP issues for which the training has been devised. Given the importance of this training, it may be appropriate to send potential trainers on a course that provides an examination certificate in HACCP principles to ensure the appropriate level of HACCP knowledge and allow the potential trainer to build confidence. It may also be useful for these supervisory staff to have some knowledge of learning styles so that they can tailor the kind of training and support they are providing to meet individual needs. They may well require support in their first training sessions, with coaching from professional trainers and feedback on the effectiveness of their first attempts at training.

13.5 Training and senior management

The senior management team clearly needs to be fully supportive of a HACCP programme if it is to be successful. Senior managers need to grasp both the importance of HACCP to the organisation and the immediate and long-term resource issues involved in implementing a HACCP programme. This is not always a straightforward task. Motivation can be a problem, particularly with SMEs where HACCP can initially be viewed as a bureaucratic burden imposed from outside, rather than as an essential tool in the management of product safety. Research by such organisations as the World Health Organisation has shown that some SMEs regard HACCP as unnecessary, expensive, complicated and time consuming to implement. There can also be problems in understanding the concept and role of HACCP, for example in relation to prerequisite systems such as good manufacturing practice (GMP) or to existing quality systems.

A key element in successful training at this level is an open discussion with full participation from all disciplines, particularly those who will need to commit the most resources to the development and implementation of the HACCP programme. Indeed this step may need to occur before the organisation becomes committed to using HACCP. The senior team must feel a sense of involvement in and ownership of HACCP, and this can only be achieved if, individually and collectively, they have had an opportunity to talk through all the issues. While it may be helpful, if an organisation is new to HACCP, to use an outside agency to open a discussion by outlining what a HACCP system is and how it works, the discussion will need a

HACCP 'champion' from within the organisation's own ranks able to relate the implementation of HACCP to the organisation's goals and needs, and to illustrate the particular benefits of a HACCP system to the organisation.

In dealing with the question of motivation, it is important to stress the practical benefits of effective HACCP implementation, not least for SMEs. These benefits include:

- focusing resources more efficiently on the key areas of safety concern
- increasing the confidence of existing customers and regulatory authorities, resulting in the need for fewer product checks or audit requirements
- compliance with a major element of some supplier quality assurance schemes, for example those of some major retailers
- improved attractiveness to other potential customers within the food industry
- demonstrating 'due diligence' in the event of any criminal or civil legal action
- improvements in the level and consistency of product quality, and in process efficiency: this is often a consequent effect of the general tightening of standards that HACCP brings.

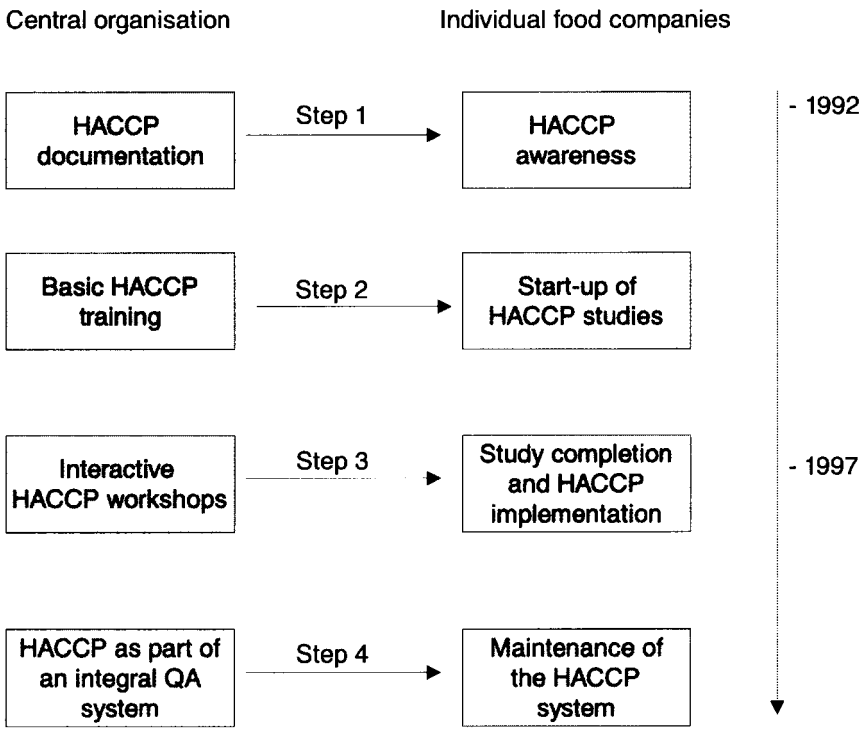


Fig. 13.2 HACCP roll-out in a multinational foods environment (source: de Winter 1998).

In explaining the nature of HACCP, a useful diagrammatic representation is shown in Fig. 13.2 where HACCP at the centre deals with significant hazards specific to the process, prerequisite programmes support HACCP by managing general issues and the quality management system effectively manages both elements in practice. Formal certification of the quality system to a standard such as ISO 9001 may be a bonus but is not a requirement. The use of models such as this helps trainees to understand how food safety and quality systems complement each other and overcomes the frequently asked question: 'Why do I need HACCP if I've already got an internationally recognised quality system?'

13.6 Training the HACCP team and team leader

Most HACCP initiatives start with the appointment of a HACCP team and HACCP team leader. Their skills will be the essential driver of the HACCP programme, so an effective training programme is essential. Responsibility for the formation and training of the team often relies on the team leader's initiative so the right training for such a role is critical. There are a number of problems in making training for both leader and team effective, including:

- ignorance of the range of skills and training required
- the resulting reliance on a one-off course for the team leader, neglecting the importance of also training other team members
- getting the timing right – training can be too early and wasted.

Of these, the first is the most serious since it will lead to an underestimation of the training needs of a HACCP team and leader. This failure will be compounded if training is undertaken too early. Before training is contemplated, it is important that the decision to implement a HACCP system within the organisation has already been taken by senior management. It is also essential that an outline schedule for HACCP implementation is agreed, in which the training of the HACCP team leader and team is the first step. If this is done, there will be a clear focus for the training and it can immediately be followed up by starting to put the training into practice.

The HACCP leader requires a range of skills. The advanced HACCP training standard mentioned in section 13.4 identifies the key requirements of any team leader training programme, and is designed to benchmark such programmes against an agreed standard of good practice. These skills are summarised in Table 13.1. In addition to specific subject knowledge, the standard also emphasises broader skills such as those of project management, team leadership and information management (in overseeing implementation of an effective and practical system of HACCP documentation).

Given the range of such skills, it is essential to select a team leader with at least some existing leadership and project management experience, and not just the right subject knowledge. It has also been suggested that it may, in some circumstances, be better to train good project managers in HACCP skills, rather

than HACCP experts in project management (see Chapter 7). Other practitioners suggest that a production/operations manager will be an effective leader due to their day-to-day requirement to get things done on schedule and within budget. This profile suggests that the team leader needs to be a relatively senior member of the organisation. By definition such people are usually the busiest in the organisation, and there may be a temptation to delegate the role to a less experienced or more junior member of staff. Given the pivotal role played by the team leader, and how demanding it is, giving in to such a temptation may compromise the entire process of HACCP implementation.

De Winter has described the training of HACCP team leaders within a multinational setting (de Winter 1998). In this model (Fig. 13.2), a central task force developed initial briefing documents and implementation software, and organised the training programme. The latter consisted of the following:

- Initial briefings on HACCP principles for QA managers and site directors to promote awareness and enthusiasm among key personnel.
- Week-long HACCP training courses for HACCP team leaders from individual companies and all the major product groupings (ice cream products, for example), with the objective of equipping them to begin the process of HACCP implementation independently.
- Once implementation was under way, workshops based on real implementation issues within a specific factory for each of the product groupings. These workshops allowed delegates to discuss common experiences and problems, share best practice and benefit from detailed advice from the central QA and microbiology group. Delegates then cascaded best practice within their own companies.
- Subsequent briefing sessions for individual product groupings to review performance and discuss changes.

He outlined an initial week-long course with a mix of lectures on HACCP issues consolidated by group exercises on HACCP analysis and planning to reinforce learning and provide a bridge to the process of implementation itself. This initial training was followed up within a year by a series of workshops for team leaders working in the same product areas. Each workshop took place in a factory, which provided a real-life case study on which the delegates worked collectively.

A similar approach has been used by the author within Cadbury Schweppes (Fig. 13.3) where an initial introductory course for team leaders and senior team members was followed up on-site by HACCP facilitation workshops involving the whole HACCP team(s). This approach allowed the site HACCP teams to learn HACCP study skills while working on their own processes. The outcome of this facilitation approach was not only effective training for HACCP team members, but also the completion of draft HACCP plans for real process modules, which could then be reviewed, validated and implemented.

A key outcome of the facilitation workshops, in addition to the draft HACCP plans and timetable for other HACCP modules, was the development of detailed

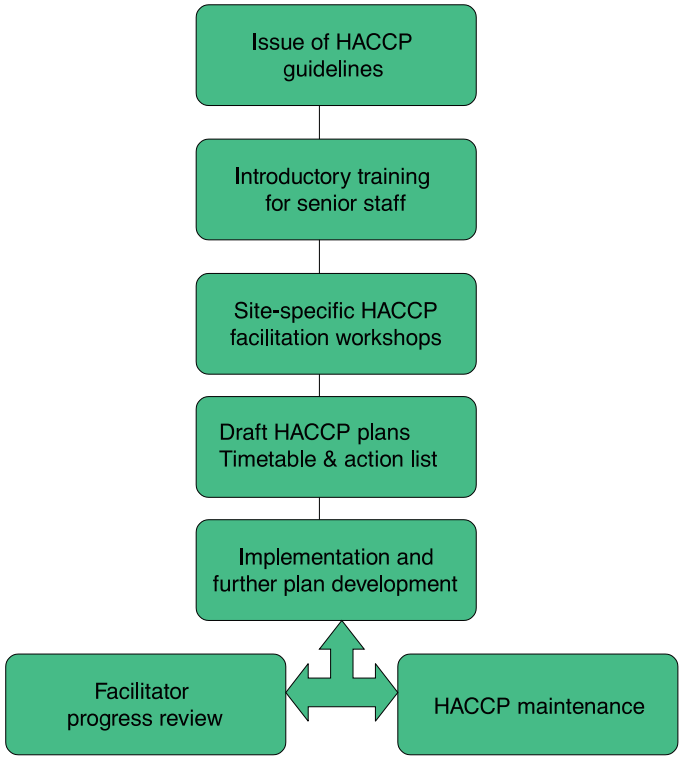


Fig. 13.3 Cadbury Schweppes International HACCP training approach.

activity lists outlining all the items to be addressed in order to implement the HACCP plans. These lists included items such as specific areas of prerequisites that required strengthening, engineering work on equipment to design out hazard access points and development of new procedures or work instructions in addition to the standard HACCP implementation activities of training and setting up monitoring/recording systems. By giving individual responsibility for each of these activities, site management was able to review and manage the progress of HACCP implementation more effectively. This contrasts with typical approaches to HACCP training either as open courses or company in-house courses, where the trainees are left with some understanding of how to do a HACCP study but are much less clear on how to move from the paperwork HACCP plan to a working HACCP system.

Following the Cadbury Schweppes facilitation workshops, the trainer was involved in ongoing progress review and encouragement of the HACCP teams through to the successful implementation of HACCP plans. This type of ongoing support and facilitation from experienced HACCP practitioners is extremely valuable but rarely seen where external trainers are used, usually due to its inherent cost.

13.7 Training CCP monitors and others

Effective monitoring of CCPs is critical to the success of any HACCP plan. CCP monitors must therefore be instructed in basic HACCP philosophy and, in particular, the range of skills involved in monitoring. They need to understand the nature of the hazards for which their CCP has been identified, how the CCP itself functions, and the nature and significance of the measurements they are taking. They need to understand the difference, for example, between a target level and a critical limit and what levels have been set for each. Finally, they must understand how to take corrective action if a deviation occurs, and to keep the appropriate records.

The training of CCP monitors is best left until the implementation phase. If carried out before then, the delay between planning and implementation will mean that much of the impact of the training may be lost, and staff may become demotivated. Such training can be difficult because the role of CCP monitors may well require new working practices and an unfamiliar level of responsibility and initiative. It might involve getting to know a particular production process in more depth, learning about unfamiliar hazards, and getting to grips with new skills in data analysis and record keeping. Employees may well be unused to taking decisions on corrective action on their own initiative. A simple briefing would be insufficient to resolve all these issues satisfactorily. One solution would be for trainers to involve CCP monitors in describing the relevant hazards and the nature of the CCP for themselves, and in developing their own working practices and instructions. A simplified process chart, based on the HACCP plan, could be used as a starting point for this process. Such an approach is valuable because it is interactive, encourages the monitor to 'own' the approach and allows the trainer to check and correct the trainee's understanding. This more interactive approach would have the additional benefit of using the employee's own experience of a process and daily working practices to make HACCP implementation more effective and user friendly. Indeed, there may be a genuine need to unpick bad working practices, poor equipment design and overcomplex working procedures highlighted as a result of the analysis of existing prerequisite programmes or during process analysis. Working with staff to identify and solve these problems should be seen as part of training.

Once again, follow-up needs to be an integral part of the training. Supervisors will need to be trained to provide appropriate support in the early stages of monitoring. They should ask CCP monitors periodically to take them through their new role, questioning them on key points and asking about suggestions for improvements, as well as sample checking CCP data for themselves to see if target levels are being effectively monitored and acted upon. There may well be a need for further training to reinforce points, as well as periodic refresher training to resharpen monitoring skills. Training needs also to be reflected in changes to job descriptions, new objectives and appraisal or bonus schemes.

Some level of HACCP training in the rest of company is also essential, especially in production areas. All production staff will need a basic

understanding of the importance of food safety, the role of a HACCP system in maximising food safety, how it relates to what they do, and the particular importance of CCP monitoring. One problem in such training is that of getting the right depth and balance of information for this broader group. One approach to such training has already been outlined (see section 13.3). Where a group may not have common production experience, an alternative approach to explaining the concept of HACCP analysis would be to start with a very basic and common process (e.g. making a cup of coffee), get the trainees to help construct a process flow diagram, brainstorm possible hazards and identify CCPs. Equipped with this basic conceptual understanding, they could then be introduced to the HACCP plan for the process or processes with which they were most familiar, and their role in effective HACCP implementation. In some cases trainees may have an inadequate grounding in good hygiene practice. If such training has already taken place, do staff need a refresher course? If so, could this be linked into HACCP training? As an example of such a course, the Royal Institute of Public Health and Hygiene (RIPHH) has developed a first certificate in food safety (1995b), which takes a risk-based approach to food hygiene including the HACCP concept.

The wrong level of training for this group can cause confusion and damage motivation. Chilton (2000) describes the lessons learnt from the provision of an initial training intervention that was 'heavy on the theory side of HACCP, but lacking on the practical implications for the employee'. The training provided was a scaled-down version of the material provided for supervisors and emphasised commitment, responsibility and change without dealing specifically with how each employee's job would be affected. The training was re-written to cover the basic questions:

- What is HACCP?
- Why is HACCP important?
- What commitment is required?
- What does the company have to do?
- What do I, as an employee, have to do?
- What happens next?

and the resulting sessions proved to be highly successful.

Answering such simple questions will help any organisation get the message across to the general workforce and this can be done by management or supervisors to give a personalised company approach. Training aids are also commercially available to help with this function such as the RSSL video, *HACCP – How the System Works* (Reading Scientific Services Ltd 1996).

13.8 Special issues in training

There are a range of problems in training within differing countries. Most obviously, language can be a potential barrier in training where the trainer and

the trainees do not speak the same first language. Translators are frequently not conversant with HACCP principles, and it is essential to check appropriate translation and definition of terms, e.g. hazard, risk, control and CCP. If possible, it is best to bring in a bilingual expert in HACCP from outside, or recruit a bilingual member of staff with relevant knowledge. Cultural differences can also be a problem. There may be differences in attitudes to the importance of food safety and differing perceptions of food safety risks. In some countries there may be differences in management style which might, for example, make junior managers and operational staff reluctant initially to act on their own initiative and become fully involved in HACCP implementation and ownership.

SMEs may have particular problems with HACCP training. They may both lack the internal resources to organise training themselves and the financial resources to use an external training provider. Government agencies can provide some training support. The World Health Organisation has held a consultation meeting and published guidelines specifically for HACCP implementation within small businesses (WHO 1999), which provides a planning framework for HACCP implementation, including training. In the UK, for example, MAFF has developed a range of general HACCP materials, training aids and packages, and has funded specific training among retail butchers. SMEs can access such materials either directly by approaching MAFF or by consulting their local Environmental Health Officer.

In the UK, MAFF has also funded food technology transfer centres with the aim of transferring knowledge and technology to SMEs at low cost. These centres, such as 'Food Knowledge and Know-how' based at Reading University, allow SMEs to gain access to expert knowledge and training in many areas including HACCP. Local universities and colleges are also often able to offer low-cost training solutions.

SMEs can also approach customers with HACCP systems for advice and support, including advice and materials on training. The Internet can also be a useful source of information on a range of HACCP issues, both as a place to find specific information and as a forum for discussion groups to share problems and experiences. The expertise of external training providers can also be exploited most economically by using them to train company staff, perhaps members of the HACCP team, as trainers for other company staff. In addition, training sessions can be combined, for instance by a joint course covering both food hygiene and HACCP.

Multinationals also present a range of problems in effective HACCP training. These are often associated with obtaining the same standards of HACCP training at different locations around the world. This can be overcome by using a centralised or centrally run training programme at least in the initial stages, but may involve the language difficulties mentioned above. Central training of local trainers and the provision of training resources in the appropriate language can be an effective solution. Another approach would be to define learning objectives for particular trainee groups and invite local training organisations to submit proposals for review.

Additional problems can be seen in businesses that have high turnover of staff. In these cases it is best to choose longer-term staff members where possible for the key HACCP team roles; however, there may still be change of personnel within the team. This will require additional training to bring new members up to speed and the remaining team members must work together to help new members understand the concepts. High staff turnover in the general workforce will almost certainly affect CCP monitoring and the underlying foundations of effective prerequisite programmes. Here it is important to provide a regular framework of HACCP/hygiene awareness and refresher training as well as detailed specific training for newly appointed CCP monitors.

Problems are also seen due to lack of experience and training for assessors of HACCP systems. This group includes regulators and enforcers, customers and third-party auditors. Some large food businesses have found themselves in the position of having to defend their HACCP plans because the auditors or assessors have less of an understanding of how to do HACCP than does the HACCP team that developed it. This can be very difficult and it is not unknown for companies to have to put in extra CCPs that are in reality only CCPs because their customers or auditors told them to do so.

Steps are being taken to address this issue in certain areas. Inspectors who are auditing on behalf of retail customers to the British Retail Consortium Technical Standard for Suppliers of Retailer Branded Food Products (British Retail Consortium 2000) are required to have a minimum standard of HACCP training of the RIPHH introductory-level Certificate in HACCP Principles or proven equivalent experience. Though not inspecting HACCP systems in the same depth as a full HACCP audit, these inspectors are important in highlighting the need to review and strengthen existing HACCP systems. There have also been moves to provide effective HACCP training for enforcement officers. In the UK, enforcers from a number of areas have recognised the need for training and in the past have attended open courses or contracted reputable training providers to train their staff. A number of large companies have also invited enforcement officials to join their in-house HACCP training. A new initiative in 2000, backed by the Food Standards Agency, has seen the development of a syllabus for a HACCP assessment course for enforcement officers (UK Food Standards Agency 2000). It is anticipated that as the enforcement officers gradually build up their experience, then the effectiveness of their assessment of HACCP systems will be improved.

13.9 Summary

Training is the single most important element when setting up a HACCP system, which provides not only the technical skills required to implement HACCP but also helps to change the attitudes of people. Training should not just be a ‘classroom’ activity, but an ongoing programme in the workplace; not just an isolated event, but a catalyst for lasting change. This will help HACCP become a part of the organisational culture.

Training needs to address not just key HACCP principles and methodologies, but common problems such as the following:

- not understanding the foundations required for HACCP, the prerequisite programmes
- where to go to remedy deficiencies in technical expertise
- ensuring the correct level of training for all staff
- failing to identify significant hazards and/or identifying too many CCPs, i.e. confusion between control points and CCPs
- not knowing what to do at the end of the HACCP study so that the HACCP plan remains a paper document on the shelf rather than becoming a living system
- developing unwieldy documentation.

There are currently no recognised international or national standards for HACCP training, although a number of groups around the world have proposed approaches and developed training objectives and standards. Clearly such recognition would help to ensure consistency of approach by training providers and help to provide mutual confidence in HACCP application for effective food safety around the world. At the time of writing, an international workshop on 'An international future for standards of HACCP training' is planned for the 2001 Noordwijk Food Safety and HACCP Forum. It is to be hoped that initiatives such as this lead the way to true harmonisation of HACCP training standards.

13.10 References

BRITISH RETAIL CONSORTIUM, 2000, *Technical Standard and Protocol for Companies Supplying Retailer Branded Food Products* (Issue 2), London.

CHILTON, J., 2000, *HACCP From the Front Line – Part 2*, Chilton Consulting Services, www.chiltonconsulting.com

DE WINTER, R.J.F., 1998, 'The role of interactive workshops in HACCP training in a multinational environment', *Food Control*, **9** (2–3): 147–9, Elsevier.

INTERNATIONAL HACCP ALLIANCE, 1996, *Basic Course Curriculum*, www.haccpalliance.org/

MORTIMORE, S.E. and SMITH, R.A., 1998, 'Standardised HACCP training assurance for food authorities', *Food Control*, **9** (2–3): 141–5, Elsevier.

MORTIMORE, S.E. and WALLACE, C.A. 1997, *A Practical Approach to HACCP: Training Programme*, Blackie Academic and Professional, London (2nd edn in press 2001 as *HACCP Trainers Resource Pack*, Aspen Publishers Inc., Gaithersburg).

MORTIMORE, S.E. and WALLACE, C.A. 1998, *HACCP: A Practical Approach*, 2nd edn, Aspen Publishers Inc., Gaithersburg.

READING SCIENTIFIC SERVICES LTD, 1996, *HACCP – How the System Works* (introductory video), Reading.

ROYAL INSTITUTE OF PUBLIC HEALTH AND HYGIENE, 1995a, *HACCP Principles and Their Application in Food Safety (Introductory Level) Training Standard*, London.

ROYAL INSTITUTE OF PUBLIC HEALTH AND HYGIENE, 1995b, *The First Certificate in Food Safety*, London.

UK FOOD STANDARDS AGENCY, 2000, *Syllabus for HACCP Assessment Course for Enforcement Officers*, London.

UK STEERING GROUP ON HACCP TRAINING STANDARDS, 1999, *HACCP Principles and Their Application in Food Safety (Advanced Level) Training Standard*, RIPHH London.

WHO, 1999, *Strategies for Implementing HACCP in Small and/or Less Developed Businesses*, Report of a WHO Consultation, WHO Food Safety Programme, Geneva (www.who.int/fsf)